Sepehr Vahedi

Software Engineer (3.5 years) | Backend, Distributed Systems, ML Infrastructure

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Right to work: available from Sep 2027 on a UK Graduate visa (18 months, no sponsorship needed to start)

EDUCATION

MSc Software Systems Engineering — Sep 2026 – Sep 2027 (expected)

University College London (UCL) — *London, UK*

- Modules: Software Architecture, Validation and Verification, Financial Institutions and Markets, Industrial Group Project.

BSc Computer Engineering (Software Engineering) — Sep 2020 – Jul 2026

Sharif University of Technology — *Tehran, Iran*

- GPA 16.71/20 (**18.25/20 in the final two years**); six years due to co-op and full-time work at Tapsi, then 2025 in research.

EXPERIENCE

Tapsi – *ride-hailing and delivery super app, 10M+ monthly active users* — *Tehran, Iran*

Senior Software Engineer, Delivery and Marketplace — Dec 2025 – Sep 2026

- Designed and owned the delivery-routing gRPC service (Kotlin, Spring WebFlux) behind every motorcycle ETA and distance estimate (ML first, then OSRM, then Haversine); migrated five services off the legacy Go estimator with no regressions.
- Raised ETA model-server throughput **54x, from 70 to 3,800 requests/sec per pod**, and cut p50 latency at peak from 866 ms to 62 ms by removing a Triton serialization bottleneck and batching sub-model calls; outputs verified identical on 19,200 requests.
- Took the delivery domain's first ML models to production on Triton (KServe v2 gRPC, CI/CD to production, accuracy gates); designed the six-city shadow and switchback A/B rollout, balanced for time of day: **ETA MAPE fell from 24% to 19%**.

Senior Software Engineer — Jan 2024 – Dec 2024

- Built the event-sourced Kafka backend (1M+ events/day) for multi-delivery orders; the launch **doubled daily deliveries**.
- Rewrote the ride-request service, which manages the lifecycle of every ride request, from a Node.js monolith to Spring Boot and Project Reactor with zero downtime, cutting its infrastructure cost **45%** and peak response time from 200 ms to 80 ms.

Software Engineer — Feb 2022 – Dec 2023

- Cut GPS-service load **67%** with adaptive driver GPS intervals, caching and filtering, and rewrote reverse geocoding in Go.
- Built the backend for a new business line, Iran's first school-transport platform (**2,000+** drivers, **12,000+** students), as one of two backend engineers on a 1.5-month deadline; designed its multi-app architecture and offline route matching.

RabinCash | Co-Founder and Product Designer — Feb 2020 – Feb 2021

- Co-founded a P2P and B2P crypto exchange (**20+ trading pairs**); owned its product and UX design from concept to launch.
- Led a two-person front-end team that built the web trading platform (JavaScript, HTML/CSS, Bootstrap).

RESEARCH EXPERIENCE

TestCouncil: Multi-Agent LLM Unit-Test Generation (BSc thesis) — Aug 2025 – Jul 2026

Software Quality Research Lab, Sharif University; supervisor: Prof. Seyed-Hassan Mirian-Hosseinabadi — *Tehran, Iran*

- Qwen, DeepSeek and Gemini agents plus an LLM synthesis step that removes ~85% of redundant tests: **84.6% line and 92.5% branch coverage** on 135 HumanEval, MBPP and algorithm functions, **36+ points** above the best single model ($p < 0.001$).

LLM Issue-Title Generation — Mar 2025 – Feb 2026

Intelligent Software Engineering Lab, Sharif University; supervisor: Prof. Abbas Heydarnoori — *Tehran, Iran*

- Cut LLM input-token cost **37.3%** via learned context selection (2.6M+ GitHub-issue dataset), within 0.004 BERTScore of GPT-4.

PROJECTS

- **Mesh-level rate limiter** (self-initiated at Tapsi): per-user, per-API token-bucket limits as an Envoy Lua filter on Istio, adopted by every microservice exposed to external clients; part of my promotion case.
- **Property platform** (Air Properties, Dubai): ETL for 10K+ listings; RAG recommender, 500+ queries/day, **40% faster matching**.
- **Multi-agent paper critique**: four LLMs cross-rank each other's critiques (shuffled against position bias), merged by Borda count.

LEADERSHIP AND MENTORING

- **Tapsi:** led the delivery-routing and ML-serving projects end to end with a data scientist, owned code review and on-call for the routing stack, and was the sole onboarding mentor through a new engineer's first three months.
- **Teaching Assistant**, Sharif: Advanced Programming (2024); Scientific and Technical Presentation (2025–26).
- **Informatics Olympiad Teacher**, NODET (2020–21): my students won **one gold and two silver INOI medals**.

SKILLS

Languages	Kotlin, Java, Python, Go, C++, TypeScript and JavaScript (Node.js), SQL
Backend	Spring Boot, Project Reactor (WebFlux), gRPC, REST, Apache Kafka, event-driven microservices, JUnit, Mockito
ML Serving	NVIDIA Triton Inference Server, KServe v2 protocol, ONNX, PyTorch, Hugging Face Transformers, LLM APIs
Infrastructure	Kubernetes, Docker, Helm, ArgoCD, GitLab CI/CD, Istio, Envoy, Prometheus, Grafana, Linux, Git
Data	PostgreSQL, MongoDB, Redis, Hazelcast, S3-compatible object storage, Uber H3, OSRM
Spoken	English (IELTS 8.0), German (intermediate), Persian (native)

AWARDS

- **Codeforces Candidate Master** (max 2075, top 1.5%); Expert (max 1765) on my earlier account; 420+ problems solved.
- **Iran National Olympiad in Informatics**, top-50 finalist (2018, 2019); **AUT International Programming Contest**, 3rd of 65.